

AHMED MOHAMMED

Machine Learning Engineer / Applied AI / Computer Vision
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PROFESSIONAL SUMMARY

ML Engineer with a completed M.Sc. in Artificial Intelligence from JKU Linz (advised by Prof. Sepp Hochreiter; thesis submitted Jul 2026, graduated Sep 2026). Applied machine learning across images, text, and time-series, with hands-on PyTorch, PyTorch Lightning, and diffusion models. Industrial delivery at PROFACTOR GmbH on the Government of Upper Austria project Zer0P: defect detection and inkjet print classification under real manufacturing constraints. Strong fundamentals in Python, Git, Docker, and Linux.

CORE SKILLS

ML Frameworks	PyTorch • PyTorch Lightning • Diffusion Models (DDPM) • CNNs • UNet • Transformers • LLMs
Research	Hydra • Experiment design • Ablation studies • AUROC/FPR95 • Cross-validation • Reproducible benchmarking
Computer Vision	OOD detection • Object detection (YOLOv8) • Defect detection • Image classification
ML Engineering	End-to-end pipelines • Training & optimization • Hyperparameter tuning • LLM inference • Prompt engineering • Structured output extraction
Software & DevOps	Python (advanced) • Git • Docker • Linux • REST APIs • CI/CD concepts • Structured logging
Web (familiar)	TypeScript • Next.js • React/Vite PWA • Supabase. Built with AI-assisted tooling; core depth is Python and applied ML.
AI Tools	Claude Code (daily) • GitHub Copilot • Microsoft 365 Copilot • ChatGPT

PROFESSIONAL EXPERIENCE

ML Researcher	Dec 2024 - Jul 2026
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JKU Institute for Machine Learning, Linz, Austria

- Thesis: *Conditional Diffusion Models as Generative Classifiers for Out-of-Distribution Detection in Inkjet Print Quality Control*, supervised by Prof. Sepp Hochreiter; assistant supervisor Claus Hofmann, MSc.
- Developed a binary conditional diffusion model with a novel class-conditional separation loss, raising OOD detection AUROC from $92.52\% \pm 11.07\%$ to $99.03\% \pm 0.07\%$ on a CIFAR-10 airplane-vs-rest split (single ID class; +6.5 pp gain, 3-seed mean).
- Achieved 98.98% AUROC on the within-CIFAR airplane-vs-rest split (seed-42) and 90.50–96.97% across five external OOD benchmarks (CIFAR-100, Places365, FashionMNIST, SVHN, Textures).
- Built modular, reproducible experiment infrastructure using PyTorch Lightning and Hydra with configuration tracking, seed-controlled runs, and auditable raw-score artefacts.

Machine Vision Researcher	Apr 2024 - Nov 2024
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PROFACTOR GmbH, Steyr, Austria (with JKU Linz)

Government of Upper Austria, project Zer0P: zero-defect inkjet printing on building components

- Designed a multi-head conditioning mechanism and a two-stage YOLOv8 then conditional-diffusion pipeline for classification of inkjet experimental prints; 8-method comparison with 5-fold cross-validation on the public FTI_Zer0P benchmark (thesis-documented baseline 0.8673 ± 0.0230 AUROC).
- Evaluated production decision thresholds in a real-time inkjet quality setting (threshold-dependent operational accuracy; distinct from the public AUROC baseline).

AI Research Intern	Aug 2023 - Oct 2023
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Karunya University, Coimbatore, India

- Developed deep learning models for EEG-based motor imagery classification in Brain-Computer Interface systems: signal preprocessing and feature extraction pipelines for multi-class EEG data, and a CNN versus RNN architecture comparison for real-time BCI decoding.

Ledzone, Karakoey / Beyoglu, Istanbul, Turkey

- PCB assembly, soldering, and component-level rework on LED and motor driver circuits; schematic and layout reading; functional test and final inspection with multimeter, oscilloscope, and function generator; ESD-aware handling and BOMs.

SELECTED PROJECTS

OOD Detection with Conditional Diffusion Models (M.Sc. Thesis) | JKU Institute for Machine Learning, Prof. Sepp Hochreiter

Binary CDM achieving $99.03\% \pm 0.07\%$ AUROC (3-seed mean) on CIFAR-10 airplane-vs-rest via a novel separation loss. Extended to industrial inkjet quality control with multi-head conditioning (two-stage YOLOv8 and CDM pipeline). Modular implementation: PyTorch Lightning, Hydra, reproducible auditable artefacts.

Sihem: LLM Personal-Mentor Assistant | personal side project, solo developer
Habits and daily-routine assistant delivered as a Telegram bot and an installable PWA. Supabase Postgres (RLS, pg_cron, pgvector, Vault secrets), Supabase Edge Functions in Deno/TypeScript, React/Vite PWA with service worker and web push, multi-provider LLM routing with escalation (Zhipu GLM primary, plus Gemini, OpenRouter, Groq, Cloudflare Workers AI), GitHub Actions deploys, German and English delivery lanes. Accepted into the tech2b incubator, Linz.

EEG Motor Imagery Classification for Brain-Computer Interfaces | Karunya University, India

Deep learning models for multi-class EEG signal classification in BCI systems. Signal preprocessing, feature extraction pipelines, and CNN/RNN architecture comparison for real-time decoding.

EDUCATION

M.Sc. Artificial Intelligence

Oct 2024 - Sep 2026

Johannes Kepler University (JKU), Linz. Thesis submitted Jul 2026, graduated Sep 2026

- Thesis advisor: Prof. Sepp Hochreiter • Assistant: Claus Hofmann, MSc • Coursework: Advanced Deep Learning, Probabilistic ML, Computer Vision, Reinforcement Learning, Statistical Machine Learning

B.Sc. Mechatronics Engineering

Feb 2015 - Jan 2018

Eastern Mediterranean University, Cyprus

- Graduation project: SCARA robotic system for dynamic object tracking, covering real-time control and sensor integration.

Electrical Engineering studies (no degree)

Oct 2011 - Jan 2015

Higher Technological Institute (HTI), Cairo, Egypt

Electrical engineering fundamentals and hands-on PCB construction. Transferred to Mechatronics Engineering at Eastern Mediterranean University; this programme was not completed.

LANGUAGES

Arabic Native

English B2, Professional Working

German B1 CEFR, currently preparing for the B2 exam

WORK AUTHORIZATION

Eligible for the Austrian **Rot-Weiss-Rot Karte for Graduates** (RWR Karte fuer Studienabsolventen), having completed my M.Sc. at JKU Linz (Sep 2026). This route requires **no AMS labour market test** and is **not subject to a quota**: an Austrian employer only needs to confirm the position and salary. There are no agency fees or sponsorship costs for the employer.